

CO2 AT3-Class Test

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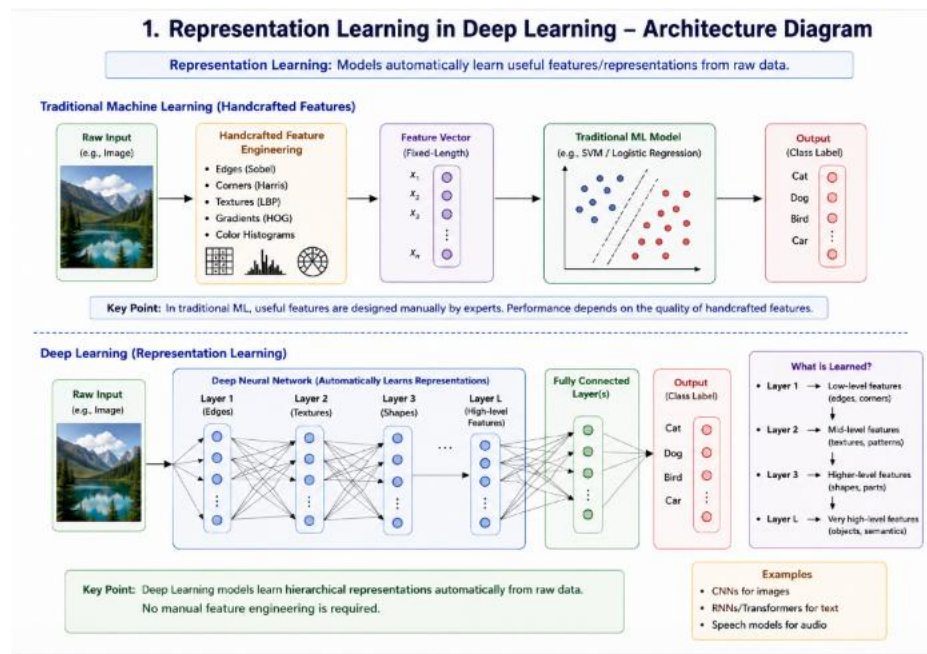
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1. Representation Learning in Deep Learning

Representation Learning is a technique where a deep learning model automatically learns useful features or representations directly from raw data. For example, in image recognition, a CNN can learn edges, textures, shapes, and finally complete objects without manually defining these features.

In traditional machine learning, humans often perform feature engineering before training the model. Deep learning reduces this requirement by learning multiple levels of representations automatically, making it especially effective for complex data such as images, speech, and text

ARCHITECTURE DIAGRAM

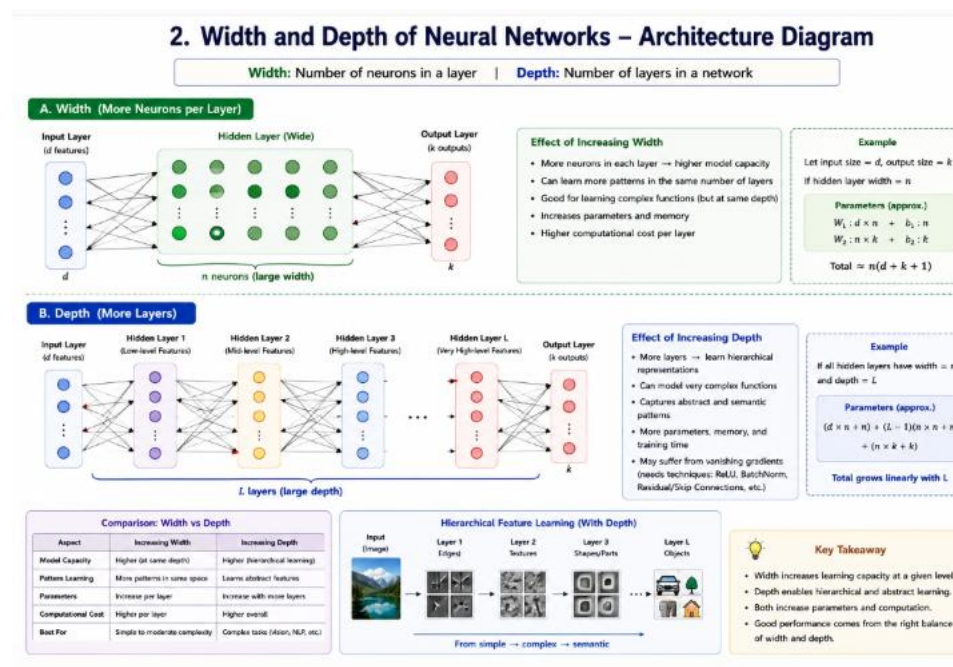


2. Width and Depth of Neural Networks

Width refers to the number of neurons in each layer, while depth refers to the number of layers in a neural network. Increasing width generally increases the model's capacity to learn more patterns, whereas increasing depth allows the network to learn more complex and hierarchical features.

However, making a network wider or deeper also increases computational cost, memory requirements, and training difficulty. Very deep networks can suffer from problems such as vanishing gradients, although techniques such as ReLU, Batch Normalization, and residual connections can help overcome them.

ARCHITECTURE DIAGRAM



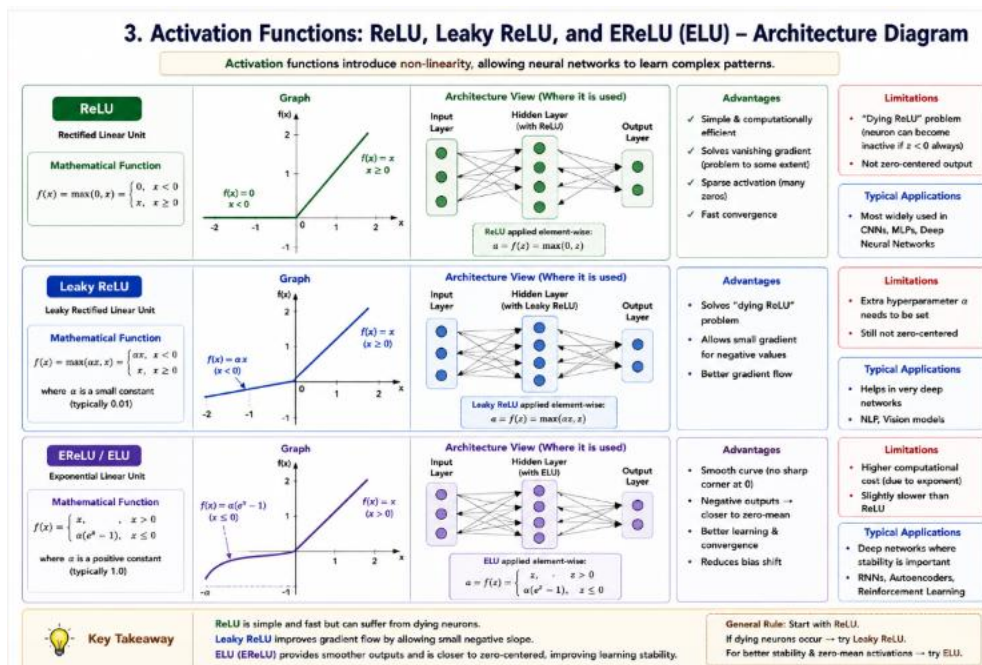
3. ReLU, Leaky ReLU and EReLU

ReLU is defined as $f(x) = \max(0, x)$, so it outputs zero for negative values and the input for positive values. Leaky ReLU allows a small negative value instead of completely becoming zero, which helps reduce the dying ReLU problem.

EReLU/ELU-style activations provide a smoother negative region and can improve learning stability.

ReLU is simple and computationally efficient, making it widely used in deep CNNs. Leaky ReLU is useful when neurons may become inactive, while EReLU/ELU is useful when smoother activation behavior and better handling of negative inputs are desired.

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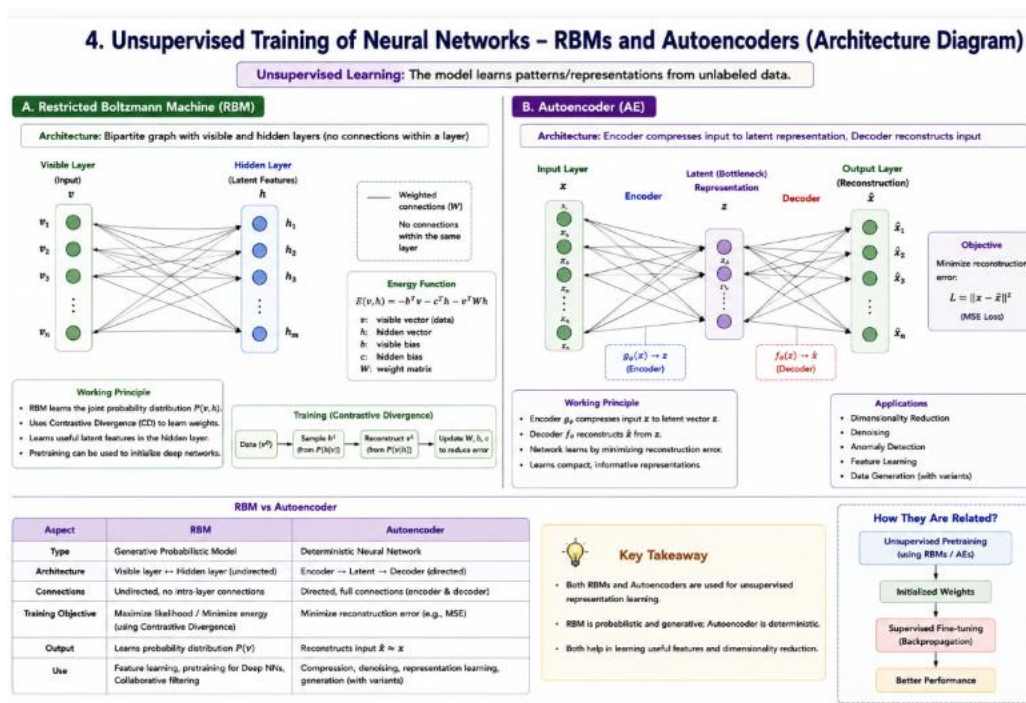


4. Unsupervised Training, RBMs and Autoencoders

Unsupervised learning trains neural networks using data without predefined output labels. The model tries to discover hidden patterns, structures, or useful representations from the input data itself.

A Restricted Boltzmann Machine (RBM) contains visible and hidden layers and learns relationships between them using probabilistic methods. An Autoencoder consists of an encoder and decoder; the encoder compresses the input into a latent representation and the decoder reconstructs the original input. Both are useful for feature learning, dimensionality reduction, and representation learning.

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5. Applications of Deep Learning

Deep learning is widely used in computer vision for image classification, object detection, and face recognition. In Natural Language Processing (NLP), it is used for translation, chatbots, sentiment analysis, and text generation.

In healthcare, deep learning helps analyze medical images and detect diseases. In speech recognition, it converts spoken language into text and enables voice assistants. In autonomous systems, deep learning helps vehicles and robots detect objects, understand their surroundings, and make decisions in real time.

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